

Solution Requirements

Date	03 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Problem Definition	Identify and define the agricultural problem that the crop recommendation system aims to solve.	High
2	Business Requirement Analysis	Gather and analyze business requirements to understand project objectives and expected outcomes.	High
3	Literature Survey	Study existing crop recommendation techniques and machine learning approaches.	Medium
4	Impact Analysis	Analyze the social and business impact of an AI-	Medium

		driven crop recommendation solution.	
5	Data Collection	Download, import and explore agricultural datasets for analysis.	High
6	Data Analysis	Perform univariate, bivariate and multivariate analysis on agricultural parameters.	High
7	Data Preprocessing	Handle missing values, outliers, feature extraction and train-test split.	High
8	Model Development	Build K-Means and Logistic Regression models, evaluate and save the best model.	High
9	Application Development	Develop HTML frontend, Flask backend and integrate the ML model.	High
10	Testing & Validation	Test the application and verify accurate crop recommendations.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance	Generate crop recommendations quickly.	Response time < 2 seconds
2	Scalability	Support future datasets and more users.	100+ concurrent users
3	Security	Protect user input and prediction data.	Secure data handling
4	Reliability	Provide consistent predictions.	Availability >95%
5	Usability	Simple and user-friendly interface.	Prediction in ≤3 steps

6	Maintainability	Modular, reusable code.	Easy deployment & updates
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